TAYLOR RILEY

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Technical leader with 15 years in tax software who builds production systems and leads the teams that ship them. Principal AI Engineer, Tax at Taxwell, building AI developer tools, a cited tax knowledge service, and the verification harnesses that decide which AI-written changes get trusted. Previously Director of Tax, owning federal tax development across two major product lines (TaxAct, Drake) from IRS regulation interpretation through calculation engine development to e-file certification. Built and deployed AI-powered development tooling—including custom MCP servers, code intelligence platforms, and automated triage systems—that fundamentally changed how the team works. Serves as the department's AI ambassador, leading evaluation and adoption of AI tooling across a regulated engineering organization.

PROFESSIONAL EXPERIENCE

Taxwell (TaxAct & Drake Tax) | Remote / Cedar Rapids, IA *2011 – Present*

Taxwell owns TaxAct and Drake Tax, serving 5M+ individual and professional tax filers. Federal Tax Development owns all federal form calculations, e-file schemas, and IRS compliance across both platforms.

Principal AI Engineer, Tax *2026 – Present*
Director, Tax *2026*
Senior Manager, Federal Tax Development *2025 – 2026*
Manager, Tax Development *2023 – 2025*

- **Organization Leadership:** Own federal tax development, delivered through two manager-led teams, responsible for 100+ federal tax forms across two codebases, two product families, and a multi-year engine consolidation initiative; built and ran the 28-person organization as Senior Manager.
- Consolidated two previously separate tax development departments into a single unified team, establishing the career ladder (Analyst I through Principal), writing all four job descriptions, and creating individual development plans for direct reports.
- Promoted a senior analyst to Federal 1040 Manager and navigated a sensitive role transition moving a team lead to a state tax principal position—both executed with zero attrition or disruption.
- **AI Platform Innovation:** Architected and personally built a code intelligence platform (Python, FastMCP, SQLite FTS5, PyQt6) that indexes 63 repositories, 660K+ calculations, and 1.5M+ cross-references into a 2 GB knowledge graph with 37 MCP tools—83 commits over 6 days.
- Created a 150-file AI documentation system that eliminated LLM fabrication of proprietary DSL syntax—previously the single largest source of wasted developer time when using AI tools. System includes compiler-extracted token validation, 16 automated checks, and a fabrication linter running in CI.
- Built two additional MCP servers (TypeScript) wrapping the internal compiler and regression test harness, closing the loop on a fully AI-assisted write-build-test-analyze development cycle.
- Serve as the department's AI ambassador; championed enterprise Claude adoption, secured pilot access for the team, and integrated AI Builder into 3 Power Automate flows for ticket triage, email classification, and daily operational recaps.
- **Data-Driven Operations:** Built 6 production Power BI dashboards from scratch (HappyFox, ADO Analytics, ATS Reporting, Federal Readiness, FormsTracker, TY25 Readiness), replacing manual spreadsheet processes with real-time self-service reporting.
- Identified and drove SLA breach rate from ~38% to ~5.6% through dashboard visibility. Surfaced workload concentration risk (top 5 analysts handling 69% of tickets) that led to redistribution and process changes.
- Designed and deployed 7 Power Automate flows automating ticket recaps, critical-issue triage, email routing, deployment notifications, and change request monitoring.
- **Platform Modernization:** Led the strategic analysis and sprint planning for the PowerBASIC-to-MathMaster DSL engine consolidation—the department’s largest multi-year initiative. Selected 1041 (Fiduciary) as the pilot product and personally converted 9+ complex IRS forms through full-lifecycle migration.
- Drove Schedule H regression diffs from initial failures to 0 diffs over 4 rounds. Identified and fixed a Form 2441 calculation bug affecting ~3,000 e-filed returns with up to $700/return in missed credits—deployed to production within 48 hours.
- Own the weekly executive briefing (VP Tax Dev, CTO, peer Directors), producing data-backed slide decks with quantified metrics and actionable recommendations—shifting the conversation from status reporting to strategic advising.

- Redesigned the Drake patching cadence from daily releases to twice-weekly, cutting context-switching by ~60% and freeing ~3–4 hours/week. Initiated cross-training to eliminate single-point-of-failure risk in the release process.
- Reduced ADO OData API calls from 100+ to ~5 per refresh by rewriting the Power Query architecture, cutting dashboard refresh time by ~80%.

Principal Tax Analyst *2011 – 2023*

Individual contributor and senior technical expert across all TaxAct product lines—individual, corporate, partnership, estate, and trust returns.

- Designed calculations, functionality, UX flows, and XML e-file schemas for TaxAct's full suite of consumer and professional tax products.
- Launched revenue-generating products including All-Inclusive Bundle, TaxAct Complete, and E-File Concierge. Owned products like TaxPlan, Refund Booster (W-4 calculator), and Tax Calculator.
- Built and deployed features with direct regulatory impact: e-file for amended returns, PayPal integration, Economic Impact Payments (EIP), and California Health Care Mandate compliance.
- Implemented fraud and security measures for IRS, state agencies, and internal systems. Maintained relationships with IRS and state departments through working groups and industry conferences.
- Analyzed user behavior and product performance using Google Analytics, Amazon Athena, Splunk, and Tableau to drive roadmap decisions.
- Contributed to hiring, training, and mentoring across the TaxAct organization, laying the foundation for the management role that followed.

TECHNICAL PROFICIENCIES

Languages & Frameworks: Python, TypeScript/Node.js, PowerShell, DAX, Power Query M, SQL, PowerBASIC, MathMaster DSL, HTML/CSS/JS

AI & LLM Tooling: Model Context Protocol (MCP) server design, Claude Code (skills, hooks, CLAUDE.md), AI Builder, prompt engineering, RAG architecture

Platforms & Infrastructure: Azure DevOps (Git, Pipelines, OData Analytics), Power BI, Power Automate, AWS S3, SQLite/FTS5, Jira, HappyFox, Snowflake, Splunk

Development Tools: FastMCP, PyQt6, ReportLab, Apryse SDK, zod, Git/SVN, VS Code

Domain: Federal tax law (individual, corporate, partnership, fiduciary, exempt), multi-state tax (CA, OH, MI, AL, DC, CO, IL, NM), IRS e-file schemas, ATS certification, tax engine architecture

COMMUNITY

VITA Volunteer — Prepared individual income tax returns for lower-income taxpayers. IRS-certified.

EDUCATION

Master of Accounting St. Ambrose University

BBA, Accounting & BBA, Finance University of Iowa